

HWAuth: Handwriting-Based Socially-Inclusive Authentication

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ABSTRACT

Small, local group of users who share private resources (e.g., families, university labs, business departments) usually have limited usable authentication needs. For these entities, existing authentication solutions either require excessive personal information (e.g., biometrics), do not distinguish each user (e.g., shared passwords), or lack security measures when the access key is compromised (e.g., physical keys). We propose an alternative solution by designing HWAuth: an inclusive group authentication system with a shared text that is uniquely identifiable for each user. Each user shares the same textual password, but individual handwriting styles of the text are used to distinguish each user. We evaluated the usability and security of our design through a user study with 30 participants. Our results suggest that (1) users who enter the same shared passwords are discernible from one another, and (2) that users were able to consistently login using HWAuth.

KEYWORDS

human computer interaction, socially-inclusive authentication

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1 INTRODUCTION

Authentication systems are typically designed for individuals with personal owned resources that has a strong motivation to keep them private (e.g., emails, bank accounts) [Cranor and Garfinkel 2005]. However, these systems are often unsuitable for a variety of small, local groups who have less-demanding security needs and collectively share information, equipment's, and/or spaces such as local organizations (e.g., a badminton club). Each member of the club should have access to this shared equipment, but members can often change so using a shared password or key can make it difficult to remove access from previous members [Bauer et al. 2007]. Gaw et al. [Gaw et al. 2006] found that using traditional individual secret for each users to access group resources could be socially inappropriate. To address these unmet requirements Das et al. [Das et al. 2017] introduced the concept of “socially-inclusive” authentication for groups who share collectively owned resources,



Figure 1: With HWAuth, users simply write the secret on a instrumental surface such as a tablet for authentication.

Traditional authentication keys are often ill-suited for socially-inclusive authentication system. On the other hand, systems that use a textual password to share amongst group of users can suffer from password theft [Zhang et al. 2010] which allows anyone with the password to be authenticated. Biometrics-based systems are less likely to suffer from theft. However, users that value privacy often worry that they will be tracked, based on their unique physiological biometrics [Science and Council 2006].

To deal with this challenging problem we explored using user's handwriting as a key for socially-inclusive authentication system. Several approaches have been proposed using handwriting for authentication [Griswold-Steiner et al. 2017; Tian et al. 2013]. In this paper we propose HWAuth and our work can be summarized as follows:

- (1) We present HWAuth, a novel socially-inclusive group authentication system that authenticates users based on group members' handwriting of a shared, textual password written on a surface instrument such as a tablet. As the secret text is shared, group members need not maintain their own individual passwords. However, because the individual handwriting of the users are unique, HWAuth can still identify individuals.
- (2) We conducted a user study to evaluate the usability and effectiveness of HWAuth and found that users who write the same shared texts are distinguishable from one another based on handwriting.

2 SYSTEM DESIGN

Figure 2 shows a high-level description of how HWAuth works. To use HWAuth, a group of users must register themselves by entering the shared secret textual password. To register, each user writes the secret text on the sensed surface numerous times (e.g. 10 times). HWAuth then processes these data from registration to train and learn each user's handwriting styles of the textual password.

HWAuth uses the model proposed by Aksan, Pece, and Hilliges in [Aksan et al. 2018] to separate style and content of the text.

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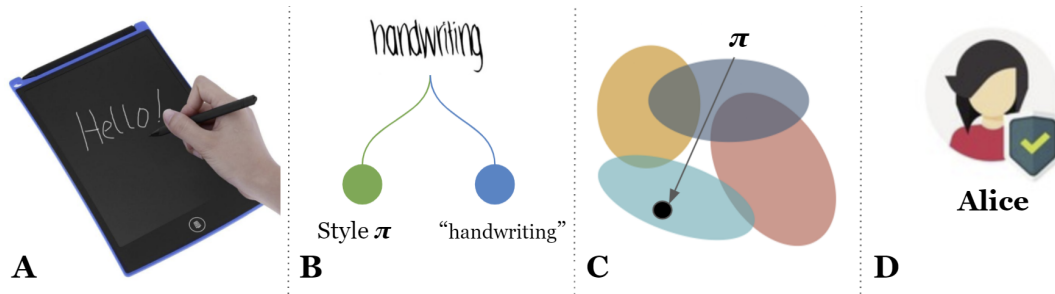


Figure 2: With HWAAuth, users write the secret password on a tablet surface (A) from which a variety of domain features are extracted to measure style π and content (B). These features are projected onto a reduced feature space, where each authentication attempt is compared against previously learned handwriting styles from users (C). If both style and content are matched, HWAAuth will authenticate the user (D).

HWAAuth treats digital ink as a sequence of temporarily ordered strokes where each text stroke consists of (u, v) pen-coordinates on the device screen and whether the corresponding pen is on (1) or off (0) the device. While the content component is defined as a set of alphabetical characters, the style component is the appearance in which these characters are formed. It is learned by a neural network (NN) architecture that contains continuous and categorical latent random variables.

To authenticate, a user writes the text in a similar manner in which they are registered. HWAAuth extracts an unlabeled feature vector from the authentication attempt and compares it against data obtained from registration. If the unlabeled feature vector meets the threshold for style and the content of the registration data, it is authenticated as the user with the highest similarity score.

3 USER STUDY

Our initial motivation for designing HWAAuth was to create a socially-inclusive authentication system that identified different users based on their handwriting with a shared textual password. To test HWAAuth, we ran a user study with 30 participants in a closed lab setting. Participants were compensated \$10 for participating in the 20 minute user study with an iPad and an Apple Pencil. Participants were informed of the purpose of the study and were given a few attempts of using HWAAuth to get familiarized with HWAAuth before the experiment began. During the experiment participants were given a textual password with the length of 8 characters to write on the iPad for registration. After 10 instances of writing for registration, participants were instructed to authenticate using HWAAuth.

As seen in Table 1, among three trials with 30 participants, 87.8% of the attempts were successful in correctly identifying the user with their handwriting.

4 CONCLUSION AND FUTURE WORK

In this paper, we designed and evaluated HWAAuth, a socially-inclusive group authentication mechanism that authenticates and identifies group members through their handwriting of a shared text. Through the user study, we found that individuals who write the same texts via tablet can be reliably distinguished from one another, and that people can write and authenticate with HWAAuth consistently. In future work, we plan to evaluate HWAAuth with different number of

Table 1: Average, median, and std. of registration and completion time in secs and average auth. success rate per trial.

Attempt	Time (sec) Avg./Med./Std.	Auth Succ. Rate
Registration	62.7/55.2/12.5	–
Auth. Trial 1	8.3/7.4/2.1	86.7%
Auth. Trial 2	8.1/7.7/1.6	86.7%
Auth. Trial 3	8.5/7.5/1.9	90.0%
Average Auth.	8.3/7.5/1.9	87.8%

character lengths for the text to find the optimal recommendations. Furthermore, we plan to implement various tests against adversaries that have access to different levels of password information to test HWAAuth’s resilience against these various attacks.

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